1. A process needs certain resources, including…, …, …, and…to accomplish its task.
Select one:

A. CPU time, memory, files, I/O devices
B. CPU time, RAM, ROM, programs
C. CPU, RAM, ROM, I/O devices

2. PLO2-PI2.1 (2) Which of the following describes primary purpose of an operating system?
Select one:

A. Primary purpose of an operating system is to allow people to use the computer.
B. Primary purpose of an operating system is to allow people to use the computer.Furthermore, it's to keep system programmers employed.
C. Primary purpose of an operating system is to make computers easier to use.
D. Primary purpose of an operating system is to provide an environment in which a user can execute programs in a convenient and efficient manner.

3. GUI stands for
Select one:

A. graphical user internet
B. graphical user interpreter
C. All are true
D. graphical user interface

4. For ensuring the efficient operation of the system itself via resource sharing, set of OS functions provides:
Select one:

A. Resource allocation, Accounting, Protection and security
B. Accounting, Protection and security, Error detection
C. A,C
D. Accounting, Protection and security

5. Choose the correct answer regarding the purpose of the system call?
Select one:

A. To request services from the operating system.
B. To interact with computer hardware
C. To provide a user interface
D. To execute user programs

6. Main memory can be viewed as…for secondary storage
Select one:

A. low-speed storage
B. fast cache
C. special storage
D. Cache

7. System Calls have…types, and they are…
Select one:

A. 5, Process control, File management, Device management, Information maintenance, Communications
B. 5, File management, Device management, Information maintenance, Communications, Memory management
C. 5, Process control, File management, Device management, Information maintenance, Information security
D. 5, Device management, Information maintenance, Communications, File Modification

8. PLO2-PI2.1 (2) Which of the following is an example of a file management system call for Windows operating system?
Select one:

A. CreatePipe(), WriteFile(), Read(), SetTimer()
B. CreateFile() , MapViewOfFile(), CreateFileMapping(), Close ()
C. CreateFile(), Read (), MapViewOfFile(), SetTimer()
D. CreateFile(), ReadFile(), WriteFile(), CloseHandle()

9. Choose the correct answer regarding the advantages of using loadable kernel modules?
Select one:

A. Difficulty in updating kernel features
B. Limited flexibility in adding new functionalities
C. Improved system performance
D. Increased kernel size

10. Choose the correct statement of process in operating system is defined incorrectly
Select one:

A. New: The process is being created.
B. Running: Instructions are being created.
C. All of the above
D. Terminated: The process has finished execution

11. Choose the state that is not a process state
Select one:

A. Blocked
B. Running
C. Exution
D. Terminated

12. PLO2-PI2.1 (2) Which of the following statement about threads is incorrect?
Select one:

A. Threads of same process share code, data and resources with one another
B. Threads have their own stack and CPU state
C. Threads slows down the processing speed due to its complex programming
D. Threads provide low cost method of implementing concurrency

13. PLO2-PI2.1 (2) Which of the following is true about program execution?
Select one:

A. Do not provides a mechanism for process communication.
B. Restrict to load a program into memory.
C. All of the above
D. Provides a mechanism for process synchronization.

14. PLO2-PI2.1 (2) In PCB,…… includes the amount of CPU and real time used, time limits, account numbers, job or process numbers
Select one:

A. I/O status information
B. Accounting information
C. CPU Scheduling Information
D. CPU registers

15. Which of the following is not an advantage of multiprogramming?
I. Increased CPU utilization
II. Better resource utilization
III. Faster response time for the user
IV. Decreased throughput
Select one:

A. IV
B. III
C. I
D. Il

16. Choose the best answer about the CPU scheduling criteria
I. Minimum: Response time, turn-around time Maximum: Processor utilization, throughput, waiting time
II. Minimum: Response time, turn-around time, waiting time Maximum: Processor utilization, throughput

III. Minimum: Response time, waiting time Maximum: Processor utilization, throughput, turn-around time

IV. All I, II, III are TRUE

Select one:

A. I
B. II
C. IV
D. III

17. Choose the correct statement of process in operating system is defined incorrectly

Select one:

A. Running: Instructions are being created
B. None of these
C. New: The process is being created.
D. All of the above

18. Choose the state that is not a process state

Select one:

A. All are TRUE
B. Blocked O
C. Terminated
D. Running

19. PLO2-PI2.1 (2) CPU scheduling is the basis of

Select one:

A. Multiprocessor systems.
B. Larger memory sized systems.
C. All of the above.
D. Multiprogramming operating systems.

20. Which of the following is not correct regarding FCFS?

Select one:

A. FCFS is easy to understand and implement. O
B. FCFS scheduling algorithm is preemptive.
C. Its implementation is based on FIFO queue.
D. Average wait time is high.

21. PLO2-PI2.1 (2) . . . scheduling is a . . . algorithm and one of the most common scheduling algorithms in batch systems.

Select one:

A. Priority/ nonpreemptive

B. Shortest Remaining Time/ nonpreemptive
C. Priority/ preemptive
D. Round Robin/ nonpreemptive

22. PLO2-PI2.1 (2) Describe disadvantage of the One to One Model ?
Select one:

A. decreased concurrency provided by this model
B. creating a user thread requires creating the corresponding kernel thread
C. creating so many threads at once can crash the system
D. increased concurrency provided by this model

23. PLO2-PI2.1 (2) which of the following is not contained by a Process Control Block (PCB)? Select one:

A. Code
B. Bootstrap program
C. Stack Od. Data

24. Which of the following is not correct regarding FCFS (First-Come, First-Served)?
Select one:

A. It may lead to the "convoy effect" where shorter processes are stuck behind longer ones.
B. It is a non-preemptive scheduling algorithm.
C. It follows the principle of first come, first served.
D. FCFS scheduling algorithm is preemptive.

25. Choose the best describe about "mutex":
Select one:

A. can be accessed from multiple processes.
B. None of these.
C. must be accessed from only one process.
D. is a binary mutex.

26. PLO2-PI2.1 (2) Select answer: Semaphore is a/an . . . to solve the critical section problem.
Select one:

A. integer variable
B. special program for a system
C. hardware for a system
D. None of these

27. Select answer: Semaphore is a/an . . . to solve the critical section problem.

Select one:

A. None of these
B. special program for a system
C. All are TRUE
D. to solve the critical

28. PLO2-PI2.1 (2) Select answer: . . . are abstract data types that provide a high-level form of process synchronization.
Select one:

A. Semaphores
B. None of the choices
C. Mutex Locks
D. Monitors

29. PLO2-PI2.1 (2) Select answer: An . . . provides uinterruptible updates on basic data types such as integers and booleans.
Select one:

A. Memory Barrier
B. Compare-and-swap
C. None of the choices
D. Atomic Variable

30. PLO2-PI2.1 (2) One disadvantage of spinlocks is:
Select one:

A. they are not sufficient for many process
B. they are too complex for programmers
C. they are unreliable sometimes
D. they require busy waiting

31. PLO2-PI2.1 (2) Choose the best describe about "mutex":
Select one:

A. can be accessed from multiple processes.
B. is a binary mutex.
C. None of these.
D. must be accessed from only one process.

32. Select answer: An . . . provides uninterruptible updates on basic data types such as integers and booleans.
Select one:

A. None of the choices
B. Memory Barrier

C. Compare-and-swap
D. Atomic Variable

33. PLO2-PI2.1 (2) Identify a system is in the safe state if:
I. The system can allocate resources to each process in some order and still avoid a deadlock.
Il. There exists a safe sequence.
Select one:

A. I
B. both I and II.
C. Neither I nor II.
D. I

34. PLO2-PI2.1 (2) Choose the best method that is a visual (mathematical) way to determine the deadlock occurrence
Select one:

A. none of the mentioned
B. resource allocation graph
C. starvation graph
D. inversion graph

35. A process is said to be in the . . . state if it was waiting for an event that will never occur
Select one:

A. Terminate
B. Blocked
C. Deadlock
D. Ready

36. Choose all that are true about the deadlock avoidance method:
I. A state is safe if the system can allocate all resources requested by all processes (up to their stated maximums) without entering a deadlock state.
II. All safe states are deadlock-free, but not all unsafe states lead to deadlocks.
III. This method does not need the system to have some additional a priori information available.
Select one:

A. I, II
B. I, II, III
C. I, III
D. II, III

37. There are ... way(s) of handling deadlocks, which are ...
I. 4, deadlock prevention, deadlock avoidance, deadlock detection, and recovery, ignore the problem altogether.

II. 3, deadlock prevention, deadlock detection, and recovery ignore the problem altogether
III. 2, deadlock prevention, deadlock detection and recovery
IV. All are TRUE
Select one:

A. III
B. I
C. II
D. IV

38. Assume that you have the following processes, their arrival times and burst times.

| Job | Arrival Time | CPU burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 3 | 7 |
| P3 | 4 | 5 |
| P4 | 11 | 4 |
| P5 | 12 | 1 |
| P6 | 15 | 3 |

For FCFS scheduling algorithms, determine the waiting times for the average waiting times.
Select one:

A. 5,3
B. 8,7
C. 9,6
D. 10

39. Assume that you have the following processes, their arrival times and burst times.

| Job | Arrival Time | CPU burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 3 | 7 |
| P3 | 4 | 5 |
| P4 | 11 | 4 |
| P5 | 12 | 1 |
| P6 | 15 | 3 |

For Round Robin (Time Quantum = 5) scheduling algorithms, determine the waiting times for P5
Select one:

A. 8
B. 7
C. 10
D. 15

40. Choose the method of data transfer that eliminates the participation of the processor during data transfer.
I. Programmed I/O
Il. Interrupt-driven I/O
III. Direct Memory Access (DMA)
IV. Input/Output Memory Management Unit (IOMMU)
Select one:

A. IV
B. I
C. III
D. II

41. Choose the wrong statement(s) about Paging Advantages
Select one:

A. Easy to use memory management algorithm.
B. Faster memory lookup times than segmentation.
C. No external fragmentation.
D. All are wrong.

42. When the valid - invalid bit is set to valid, it means that the associated page . . .
Select one:

A. is the system's physical address space
B. is in the process's logical address space
C. is in the TLB
D. has data in it

43. Apply the page replacement is "LRU" method, if all page frames are initially empty, and a process is allocated 4 page frames in real memory and references its pages in the order 1 2 3 4 12 0 1 2 3 4 5, the total number of page faults caused by the process will be
Select one:

A. 7
B. 8
C. 9
D. 5

44. PLO2-PI2.1 (2) Choose the correct statement(s) about I/O hardware
Select one:

A. I/O hardware contains: port.
B. I/O hardware contains: bus, controller.
C. I/O hardware contains: port, bus, controller
D. I/O hardware contains: port, bus, controller, devices I/O.

45. Apply the page replacement is "Optimal" method, if all page frames are initially empty, and a process is allocated 4 page frames in real memory and references its pages in the order 1 2 3 4 12 5 1 2 3 4 5, the total number of page faults caused by the process will be
Select one:

A. 6
B. 8
C. 5
D. 7

46. PLO2-PI2.1 (2) Choose the wrong statement(s) about the advantages of placing functionality in a device controller, rather than in the kernel
Select one:

A. Bugs are easy to fix-a new firmware version or new hardware is needed.
B. All of the above.
C. Performance can be improved by utilizing dedicated hardware and hard-coded algorithms
D. Bugs are less likely to cause an operating system crash

47. PLO2-PI2.1 (2) Choose the correct statement(s) about Polling Controller sets . . . . , executes transfer.
Select one:

A. wait bit
B. busy bit
C. error bit
D. command-ready

48. Choose the wrong statement(s) about the advantages of placing functionality in a device controller, rather than in the kernel
Select one:

A. Bugs are less likely to cause an operating system crash
B. All of the above.
C. The kernel is simplified by moving algorithms out of it
D. Performance can be improved by utilizing dedicated hardware and hard-coded algorithms

49. Choose the best answer:
. . . . . allocates the largest hole (free fragment) available in the memory
I. Best Fit

II. First Fit

III. Worst Fit

IV. All are TRUE
Select one:

A. I
B. IV
C. II
D. III